

a Input differential metabolites

Status	Compound name	KEGG ID	Bucket
[GT signal]	Dehydroepiandrosterone	C01227	central_metabolism
	Indoleacetaldehyde	C00637	amino_acid_metabolism
[GT signal]	Androstenedione	C00280	lipid_metabolism
	Menadione	C05377	other
	(S)-3-Hydroxyisobutyric acid	C06001	amino_acid_metabolism
[GT signal]	Estradiol	C00951	lipid_metabolism
[GT signal]	Estrone	C00468	lipid_metabolism
[GT signal]	Testosterone	C00535	lipid_metabolism

b LLM narrative · claude-opus-4-7 · 16.3s · 2,828 chars

Metabolomics Pathway Analysis

1. Most Likely Affected Pathways

The dominant signal is **steroid hormone biosynthesis** (KEGG map00140). Six of the eight metabolites sit squarely in this pathway: dehydroepiandrosterone (DHEA), androstenedione, testosterone, estrone, and estradiol form a connected biosynthetic chain. Secondary signals include:

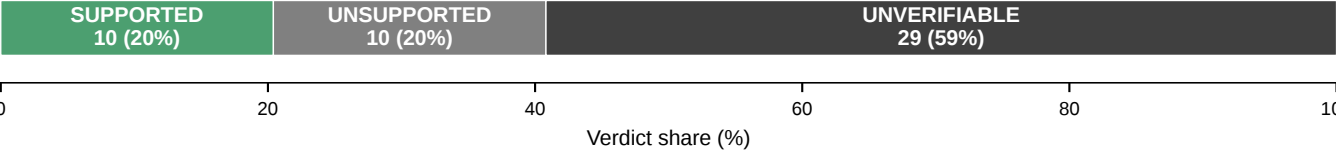
- **Tryptophan metabolism** (indoleacetaldehyde, an intermediate between tryptamine/indole-3-pyruvate and indole-3-acetate)
- **Valine degradation / branched-chain amino acid (BCAA) catabolism** ((S)-3-hydroxyisobutyric acid is a valine-specific intermediate)
- **Ubiquinone and other terpenoid-quinone biosynthesis / vitamin K cycle** (menadione, vitamin K3)

2. Key Drivers

Within steroidogenesis, the key drivers are **DHEA** and **androstenedio**

... [truncated; full text in end_to_end_case.json]

c Verifier verdict distribution



d Sample claims and verdicts (6 of 49 extracted; 2 per category)

SUPPORTED
claim #2

Steroid hormone biosynthesis maps to KEGG map00140
Evidence: Pathway ID 'map00140' resolves to RAMP_P_000000074 in RaMP, but is not in this task's enrichment context. SUPPORTED on membership, not on relevance.

SUPPORTED
claim #4

Dehydroepiandrosterone (DHEA) is one of the six metabolites in the steroid hormone biosynthesis pathway
Evidence: RaMP analytehaspathway confirms compound (RAMP_C_000000054, derived from cpd:C01227) is in at least one of the 1 resolved pathways for the claim (1 membership row(s)).

UNSUPPORTED
claim #1

The dominant signal is steroid hormone biosynthesis
Evidence: Pathway 'dominant signal is steroid hormone biosynthesis' not present in this task's ground_truth_pathway, top_pathways[:10], or RaMP. Treated as unsupported (no positive evidence) rather than contrad.

UNSUPPORTED
claim #3

Six of the eight metabolites sit in the steroid hormone biosynthesis pathway
Evidence: Pathway 'sit in the steroid hormone biosynthesis' not present in this task's ground_truth_pathway, top_pathways[:10], or RaMP. Treated as unsupported (no positive evidence) rather than contradicted.

UNVERIFIABLE

DHEA, androstenedione, testosterone, estrone, and estradiol form a connected biosynthetic chain
Evidence: Layer biological_sub6 found no pathway phrase / ID and no concrete entity to verify. Free-text biological role claims are out of v0 scope.

UNVERIFIABLE

Indoleacetaldehyde is an intermediate between tryptamine/indole-3-pyruvate and indole-3-acetate
Evidence: Layer biological_sub6 found no pathway phrase / ID and no concrete entity to verify. Free-text biological role claims are out of v0 scope.